

LECTURE-14

Gene Editing by CRISPR: Disrupting the Gene (Knock-out)

This section explains the mechanism used to deliberately disrupt a gene's function, often resulting in a **gene knock-out**.

- **Mechanism:** After the **CRISPR-Cas9** complex cleaves the target DNA, it leaves a **Double-stranded break** in the **Gene**¹.
- **Repair Pathway:** When the cell repairs this break **without external information**²²², it primarily uses a pathway called **Non-Homologous End Joining (NHEJ)**³³³³.
 - NHEJ is an "End joining" mechanism⁴.
 - It is an error-prone process where the two broken DNA ends are simply ligated back together⁵.
- **Result (Disruption):** During this repair, a few bases can get **added (Random insertions)** or **removed (Random deletions)** at the cut site⁶⁶⁶.
 - This alters the original genetic information in a **disruptive manner**⁷.
 - This often leads to a **frameshift mutation**, rendering the gene non-functional, thereby achieving a **gene knock-out**⁸.

Gene Editing by CRISPR: Repairing the Gene (Knock-in/Correction)

This section describes the method used to precisely correct or insert new genetic material, a process often referred to as a **knock-in**.

- **Mechanism:** The **DNA breaks caused by Cas9** are repaired with the aid of **external information**⁹.
- **Repair Pathway:** This pathway is called **Homology-Directed Repair (HDR)**¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰¹⁰.

- It utilizes a **Template**¹¹.
- This pathway **requires a new piece of DNA** to be introduced to the cell¹².
- **Result (Correction/Insertion):**
 - The sequence of this new piece of DNA is the **"correct" one**¹³.
 - The new DNA sequence is used as a **template to recover the information lost** when Cas9 cut the DNA¹⁴.
 - This process **allows bad information on the DNA to be corrected**¹⁵.
 - If Cas9 cuts at two places on the DNA, the **entire DNA segment in between the cuts can be fully replaced**¹⁶, enabling the **Insertion of a DNA sequence**¹⁷.

Bioinformatics Tools for Guide RNA Design

This section focuses on using online tools to design the specific **guide RNAs** needed to perform CRISPR editing on a target gene.

Finding a Target Gene

- To start gene editing research, one must first identify the target gene. An example used is the **HBB gene**¹⁸ (which codes for β -globin, mutations in which cause Sickle Cell Disease and β -thalassemia).
- Databases like **Ensembl** are used to understand the genomic location and structure of human genes¹⁹.

Using Guide RNA Design Tools (CHOPCHOP and Synthego)

- **CHOPCHOP** and other tools (like Synthego's **Knockout Guide Design**) allow researchers to **design guideRNAs for their genes of interest**²⁰²⁰²⁰²⁰²⁰²⁰²⁰²⁰.
- These tools take input parameters:
 - **Target:** The gene of interest (e.g., **HBB**²¹²¹²¹²¹).
 - **Genome:** The species (e.g., **Homo sapiens**²²²²²²²²).

- **Nuclease:** The Cas protein being used (e.g., **CRISPR/Cas9** or **SpCas9** 23232323).
- **Goal:** The desired editing outcome (e.g., **knock-out**, knock-down, **knock-in** 24242424).
- **Output:** The tools generate a list of potential guide RNAs, often with scoring metrics:
 - The results for the HBB gene show a **Rank** and the **Target sequence** 252525252525.
 - Metrics include **GC content**, **Self-complementarity**, and predicted **MM (mismatch) counts** 26262626.
 - Crucially, they assess **Efficiency** and show which guides are in an **Early Coding Region** or **Common Exon** and have **Minimal Off Targets** 27272727272727272727272727272727.
 - The goal is to select a guide RNA that is **highly efficient** and has **minimal off-target effects** to ensure specific editing.